中山大学软件学院 **2009** 级软件工程专业(2009 秋季学期)

《线性代数》期末试题试卷(B)

(考试形式：闭卷　　考试时间：2 小时)

警示

《中山大学授予学士学位工作细则》第六条

考试作弊不授予学士学位

方向：_____　　姓名：_____　　学号：_____

1. Fill the blank (4 titles * 4 points/title = 16 points)

(1) The matrics A and B below are row equivalent.

$$A=\begin{bmatrix}-2 & -5 & 8 & 0 & -17\\ 1 & 3 & -5 & 1 & 5\\ 3 & 11 & -19 & 7 & 1\\ 1 & 7 & -13 & 5 & -3\end{bmatrix};\quad B=\begin{bmatrix}1 & 3 & -5 & 1 & 5\\ 0 & 1 & -2 & 2 & -7\\ 0 & 0 & 0 & -4 & 20\\ 0 & 0 & 0 & 0 & 0\end{bmatrix}$$

A basis for $Row\ A$ is _____.

(2) If $A=\begin{bmatrix}1 & -1 & -3 & 0\\ 0 & 1 & 5 & 4\\ -1 & 2 & 8 & 5\\ 3 & -1 & -2 & 3\end{bmatrix}$, then $\det A=$_____.

(3) Find matrix $A=$_____such that $ColA=\left\{\begin{bmatrix}2s+3t\\ r+s-2t\\ 4r+s\\ 3r-s-t\end{bmatrix}: r,s,t\in R\right\}$.

(4) If α_1 and α_2 are orthonomal vectors，and $x=\alpha_1+5\alpha_2$，$y=4\alpha_1-3\alpha_2$，then $x\cdot y=$_____。

2. Mark each statement True or False, and descript your reasons (3titles * 8 points/title = 24 points)

(1) If $AB=C$ and C has 5 columns, then A has 5 columns.

(2) All polynomials of degree at most 5 consists of a vector space.

(3) Let $A\in R^{n\times n}$ and $\det(A)=a$，then $\det(3A)=3a$.

(4) If $A^T = A$ and if vectors u and v satisfy $Au = 3u$ and $Av = 4v$, then $u \cdot v = 0$.

(5) Let $A \in R^{5\times 4}$ and $rankA = 3$, then $\dim NulA = 3$.

(6) $A \in R^{n\times n}$ is invertible if and only if all columns of A are linear independent.

(7) $Nul\ A = \{0\}$ if and only if the linear transformation $x \mapsto Ax$ is one to one.

(8) If $A \in R^{n\times n}$ is diagonalizable, then A has n distinct eigenvalues

3. Calculation issues (40 points)

(1) Make a change of variable, $x = Py$, that transforms the following quadratic form into one with no cross-product term. Give P and the new quadratic form. (12 points).

$$3x_1^2 + 6x_2^2 + 3x_3^2 - 4x_1x_2 + 8x_1x_3 + 4x_2x_3$$

(2) Let the system equations be $\begin{cases} (2-\lambda)x_1 + 2x_2 - 2x_3 = 1 \\ 2x_1 + (5-\lambda)x_2 - 4x_3 = 2 \\ -2x_1 - 4x_2 + (5-\lambda)x_3 = -\lambda - 1 \end{cases}$. Find the approriate values for λ to make the system have at most one solution, no solution, and infinite solutions, respectivly. When the system has infinite solutions, write them in parametric vector form. (12 points).

(3) Find a least-squares solution of Ax = b, and compute the least-squares error. (10 points).

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix}, \mathrm{b} = \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix}$$

(4) Let $\mathfrak{B} = \{\mathrm{b}_1, \mathrm{b}_2, \mathrm{b}_3\}$ be a basis for a vector space V, and let T: V--->V be a linear transformation with the property that $T(\mathrm{b}_1) = \mathrm{b}_1 + \mathrm{b}_3, T(\mathrm{b}_2) = 2\mathrm{b}_1 - \mathrm{b}_3$, and $T(\mathrm{b}_3) = 3\mathrm{b}_1 + 4\mathrm{b}_2 + 5\mathrm{b}_3$. Find the matrix of T relative to the basis (6 points).

4. Prove issues (20 points)

(1) Let $\{\xi_1 \quad \xi_2 \quad \xi_3\}$ be a basis for R^3, and $\alpha_1 = \xi_1 + \xi_2 - 2\xi_3$, $\alpha_2 = \xi_1 - \xi_2 - \xi_3$, $\alpha_3 = \xi_1 + \xi_3$, $\beta = 6\xi_1 - \xi_2 - \xi_3$. Prove that $\{\alpha_1 \quad \alpha_2 \quad \alpha_3\}$ is also a basis for R^3, and find the coordinate vector of β relative to $\{\alpha_1 \quad \alpha_2 \quad \alpha_3\}$. (10 points)

(2) Suppose A is a $m \times n$ matrix such that the matrix $A^T A$ is invertible. Let b be any vector in R^n. Show that the Linear system $Ax = b$ has at most one solution. (10 points)

中山大学软件学院 **2009** 级软件工程专业(2009 秋季学期)

《线性代数》期末试题答案(B)

(考试形式：闭卷　　考试时间：2 小时)

警示

《中山大学授予学士学位工作细则》第六条

考试作弊不授予学士学位

方向：_____　　姓名：_____　　学号：_____

1. Fill the blank (4 titles * 4 points/title = 16 points)

(1) The matrics A and B below are row equivalent.

$$A=\begin{bmatrix}-2&-5&8&0&-17\\1&3&-5&1&5\\3&11&-19&7&1\\1&7&-13&5&-3\end{bmatrix};\quad B=\begin{bmatrix}1&3&-5&1&5\\0&1&-2&2&-7\\0&0&0&-4&20\\0&0&0&0&0\end{bmatrix}$$

A basis for $Row\ A$ is

$$\underline{\{(1\ \ 3\ \ -5\ \ 1\ \ 5),(0\ \ 1\ \ -2\ \ 2\ \ -7),(0\ \ 0\ \ 0\ \ -4\ \ 20)\}}.$$

(2) If $A=\begin{bmatrix}1&-1&-3&0\\0&1&5&4\\-1&2&8&5\\3&-1&-2&3\end{bmatrix}$, then $\det A=$ **3**.

(3) Find matrix $A=\begin{bmatrix}0&2&3\\1&1&-2\\4&1&0\\3&-1&-1\end{bmatrix}$ such that $ColA=\left\{\begin{bmatrix}2s+3t\\r+s-2t\\4r+s\\3r-s-t\end{bmatrix}:r,s,t\in R\right\}$.

(4) If α_1 and α_2 are orthonomal vectors，and $x=\alpha_1+5\alpha_2$，$y=4\alpha_1-3\alpha_2$，then $x\cdot y=$ **-11**。

2. Mark each statement True or False, and descript your reasons (3titles * 8 points/title = 24 points)

(1) If $AB=C$ and C has 5 columns, then A has 5 columns. **(False)**

(2) All polynomials of degree at most 5 consists of a vector space. **(True)**

(3) Let $A \in R^{n\times n}$ and $\det(A) = a$, then $\det(3A) = 3a$. **(False)**

(4) If $A^T = A$ and if vectors u and v satisfy $Au = 3u$ and $Av = 4v$, then $u \cdot v = 0$. **(True)**

(5) Let $A \in R^{5\times 4}$ and $rankA = 3$, then $\dim NulA = 3$. **(False)**

(6) $A \in R^{n\times n}$ is invertible if and only if all columns of A are linear independent. **(True)**

(7) $Nul\ A = \{0\}$ if and only if the linear transformation $x \mapsto Ax$ is one to one. **(True)**

(8) If $A \in R^{n\times n}$ is diagonalizable, then A has n distinct eigenvalues **(False)**

3. Calculation issues (40 points)

(1) Make a change of variable, $x = Py$, that transforms the following quadratic form into one with no cross-product term. Give P and the new quadratic form. (12 points).

$$3x_1^2 + 6x_2^2 + 3x_3^2 - 4x_1x_2 + 8x_1x_3 + 4x_2x_3$$

Solution: The matrix of the given quadratic form is $A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$, whose eigenvales are 7, 7, -2. The transformation matrix P is formed by the orthonomal eigenvectors corresponding to these eigenvalues. That is

$$P = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{18} & -2/3 \\ 0 & 4/\sqrt{18} & -1/3 \\ 1/\sqrt{2} & 1/\sqrt{18} & 2/3 \end{bmatrix}$$

The new quadratic form with no cross-product term is $7x_1^2 + 7x_2^2 - 2x_3^2$.

(2) Let the system equations be $\begin{cases} (2-\lambda)x_1 + 2x_2 - 2x_3 = 1 \\ 2x_1 + (5-\lambda)x_2 - 4x_3 = 2 \\ -2x_1 - 4x_2 + (5-\lambda)x_3 = -\lambda - 1 \end{cases}$. Find the approriate values for λ to make the system have at most one solution, no solution, and infinite solutions, respectivly. When the system has infinite solutions, write them in parametric vector form. (12 points).

Solution: The system has at most one solution as $\lambda \neq 1$ and $\lambda \neq 10$, no solution as $\lambda = 10$, and infinite solutions as $\lambda = 1$. The infinite solutions of the system can be written as $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $c_1, c_2 \in R$.

(3) Find a least-squares solution of Ax = b, and compute the least-squares error. (10 points).

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix}, \mathrm{b} = \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix}$$

Solution: The least-squares solution of Ax = b and the least-squares error are given as $\hat{x} = \begin{bmatrix} 10 \\ -6 \\ 2 \end{bmatrix}$ and $\|b - A\hat{x}\| = \sqrt{(-1)^2 + (-1)^2 + 1^2 + 1^2} = 2$, respectively.

(4) Let $\mathfrak{B} = \{\mathrm{b}_1, \mathrm{b}_2, \mathrm{b}_3\}$ be a basis for a vector space V, and let T: V--->V be a linear transformation with the property tha $T(\mathrm{b}_1) = \mathrm{b}_1 + \mathrm{b}_3, T(\mathrm{b}_2) = 2\mathrm{b}_1 - \mathrm{b}_3$, and. $T(\mathrm{b}_3) = 3\mathrm{b}_1 + 4\mathrm{b}_2 + 5\mathrm{b}_3$. Find the matrix of T relative to the basis (6 points).

Solution:

$T(\mathrm{b}_1) = \mathrm{b}_2 + \mathrm{b}_3, \quad T(\mathrm{b}_2) = 2\mathrm{b}_1 - \mathrm{b}_3, \quad T(\mathrm{b}_3) = 3\mathrm{b}_1 + 4\mathrm{b}_2 + 5\mathrm{b}_3$

Set up coordinate vectors:

$$[T(\mathrm{b}_1)]_{\mathfrak{B}} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad [T(\mathrm{b}_2)]_{\mathfrak{B}} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \quad [T(\mathrm{b}_3)]_{\mathfrak{B}} = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}. \quad \text{Set up} \quad [T]_{\mathfrak{B}} = \begin{bmatrix} 0 & 2 & 3 \\ 1 & 0 & 4 \\ 1 & -1 & 5 \end{bmatrix}$$

4. Prove issues (20 points)

(1) Let $\{\xi_1 \quad \xi_2 \quad \xi_3\}$ be a basis for R^3, and $\alpha_1 = \xi_1 + \xi_2 - 2\xi_3$, $\alpha_2 = \xi_1 - \xi_2 - \xi_3$, $\alpha_3 = \xi_1 + \xi_3$, $\beta = 6\xi_1 - \xi_2 - \xi_3$. Prove that $\{\alpha_1 \quad \alpha_2 \quad \alpha_3\}$ is also a basis for R^3, and find the coordinate vector of β relative to $\{\alpha_1 \quad \alpha_2 \quad \alpha_3\}$. (10 points)

Proof： We have $(\alpha_1 \quad \alpha_2 \quad \alpha_3)=(\xi_1 \quad \xi_2 \quad \xi_3)\begin{pmatrix}1 & 1 & 1\\ 1 & -1 & 0\\ -2 & -1 & 1\end{pmatrix}=(\xi_1 \quad \xi_2 \quad \xi_3)A$.

ξ_1,ξ_2,ξ_3 are linear independent，and $|A|=-5\neq 0$，therefore $\alpha_1,\alpha_2,\alpha_3$ are also linear independent， and consist of a basis of R^3. Let $\beta=k_1\alpha_1+k_2\alpha_2+k_3\alpha_3$. We obtain the following linear system equations

$\begin{cases} k_1+k_2+k_3=6 \\ k_1-k_2=-1 \\ -2k_1-k_2+k_3=-1 \end{cases}$. A solution to the equations is $k_1=1$，$k_2=2$，$k_3=3$. Thus, the coordinate vector of β relative to $\{\alpha_1 \quad \alpha_2 \quad \alpha_3\}$ is $\begin{pmatrix}1\\2\\3\end{pmatrix}$.

(2) Suppose A is a $m\times n$ matrix such that the matrix A^TA is invertible. Let b be any vector in R^n. Show that the Linear system $Ax=b$ has at most one solution. (10 points)

Proof：

Suppose Ax = b has two solutions. Let these solytions be x1 and x2. Thus Ax1=b and Ax2= b.Therefore Ax1 = Ax2. Mutiplying by A^T gives $A^TAx1 = A^TAx2$. So $(A^TA)^{-1}A^TAx1 = (A^TA)^{-1}A^TAx2$. So Ix1 = Ix2. Since $(A^TA)^{-1}A^TA = I$ is the identity matrix. So x1 = x2, as desired.

中山大学软件学院 2010 级软件工程专业(2010学年秋季学期)

《SE-103 线性代数》期末试题参考答案(B)

1.

(1)10; **(2)** $\begin{pmatrix}-2\\2\\-3\end{pmatrix}, \begin{pmatrix}4\\-6\\8\end{pmatrix}$; **(3)** $-\lambda^3+4\lambda^2-9\lambda-6$; **(4)** $\begin{pmatrix}-2\\1\end{pmatrix}$;

(5) $\begin{pmatrix}x_1 & x_2 & x_3\end{pmatrix}\begin{pmatrix}5 & -3 & 0\\-3 & 3 & 4\\0 & 4 & 2\end{pmatrix}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}$

2. (1)错误，当 m=n 时才是一对一的映射。
(2)错误，因为矩阵的行阶梯形式并不唯一。
(3)错误，因为例如令 A 和 B 为两个不同的 2 阶置换矩阵，则（A+B）不可逆。
(4)正确，因为两者有相同的特征多项式。
(5)错误，例如单位矩阵是对称的，但是没有不同的特征值。

3.
(1)（过程正确 5 分，计算正确 2 分）

S 的面积是 $\left|\det\begin{pmatrix}1 & 5\\3 & 1\end{pmatrix}\right|=14$**, det A=2, 所以根据第 3 章定理 10 可知，线性变换后 S 的面积是 2×14=28.**

(2)（各个基 3 分，其中过程正确 2 分，结果正确 1 分）
根据题意可知，有 T(1)=3+5t, T(t)=-2t+4t^2, T(t^2)=t^2, 所以它对应以 B 为底的变换矩阵为

$$\begin{pmatrix}3 & 0 & 0\\5 & -2 & 0\\0 & 4 & 1\end{pmatrix}$$

(3)（特征值求解 6 分，其中每个特征值 1 分，对应特征向量 2 分；矩阵对角化分解 2 分；最后计算结果 2 分，没有化简则只给 1 分）
根据题意可知，由于原矩阵是下三角阵，所以其特征值为主对角线上的元素，即 a 和 b。

代入后分别计算得其对应特征向量为 $v_a=\begin{pmatrix}1\\3\end{pmatrix}, v_b=\begin{pmatrix}0\\1\end{pmatrix}$**,所以原矩阵可以对角化为**

$$\begin{pmatrix} a & 0 \\ 3(a-b) & b \end{pmatrix} = PDP^{-1} = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix}$$

所以有

$$\begin{pmatrix} a & 0 \\ 3(a-b) & b \end{pmatrix}^k = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}\begin{pmatrix} a^k & 0 \\ 0 & b^k \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} a^k & 0 \\ 3(a^k-b^k) & b^k \end{pmatrix}$$

(4) (计算过程正确 5 分，计算结果正确 2 分)

根据最小二乘法可知，方程 $A\vec{x} = \vec{b}$ 的最小二乘解满足方程

$$A^T A\vec{x} = A^T\vec{b}$$

又因为

$$A^T A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix} \qquad A^T\vec{b} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 11 \end{pmatrix}$$

解得 $x_1 = 0.9$， $x_2 = 0.4$，即为所求方程的最小二乘解。

4. (1)（充分性与必要性证明各占 5 分）

证：记线性变换 T 的标准矩阵为 A，则由于 T 是 R^n 到 R^n 的满射，所以 A 可逆是其充分必要条件。根据第二章定理 9 可知，线性变换 T 可逆也是 A 可逆充分必要条件。

(2)（判断 3 分，理由说明 6 分）

解：一定有解。因为记原线性方程组的系数矩阵为 A，则由题意可知 A 是一个 5×6 的矩阵且原线性方程组的非平凡解中只有一个变量非零；所以 dim Nul A = 1。因此根据秩定理可知 dim Col A = rank A = 6 − 1 = 5，而 R^5 中只有它本身作为子集时维数才是 5。这表示 Col A 就是 R^5，即 A 为系数矩阵的任意一个线性方程组都是相容的。

5.证：

记矩阵 A 的特征值 λ 对应的特征向量是 $\vec{v}$，则根据定义可知 $A\vec{v} = \lambda\vec{v}$，所以有

$$p(A)\vec{v} = (c_0 + c_1A + \cdots + c_nA^n)\vec{v} = (c_0 + c_1\lambda + \cdots + c_n\lambda^n)\vec{v} = p(\lambda)\vec{v}$$

这表示 $p(\lambda)$ 就是 $p(A)$ 的一个特征值。

中山大学软件学院 2010 级软件工程专业(2010 学年秋季学期)

《SE-103 线性代数》期末试题(B 卷)

(考试形式：闭 卷 考试时间：2 小时)

警示

《中山大学授予学士学位工作细则》第六条

考试作弊不授予学士学位

方向：_____ 姓名：_____ 学号：_____

1. Fill the blank (5 titles * 4 points/title = 20 points)

(1) The determinant of $\begin{vmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{vmatrix}$ is_____.

(2) Assume that $A = \begin{pmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{pmatrix}$ is row equivalent to $\begin{pmatrix} 1 & 0 & 6 & 5 \\ 0 & 2 & 5 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$. A basis for Col A is _____.

(3) The characteristic polynomial of $\begin{pmatrix} 1 & 0 & -1 \\ 2 & 3 & -1 \\ 0 & 6 & 0 \end{pmatrix}$ is _____.

(4) The orthogonal projection of $\begin{pmatrix} 1 \\ 7 \end{pmatrix}$ onto the line through $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$ and the origin is _____.

(5) For $\vec{x}$ in $\mathbf{R}^3$, let $Q(\vec{x}) = 5x_1^2 + 3x_2^2 + 2x_3^2 - 6x_1x_2 + 8x_2x_3$. Then the matrix of the quadratic form is _____.

2. Mark each statement True or False, and descript your reasons (5titles*4pts/title=20pts)

(1) If A is $m\times n$ and rank $A=m$, then the linear transformation $x \mapsto Ax$ is one-to-one.

(2) If matrices A and B are row equivalent, they have the same echelon form.

(3) If matrices A and B are both $n\times n$ invertible matrices, then $(A+B)$ is also invertible.

(4) The matrices A and A^T have the same eigenvalues, counting multiplicities.

(5) An $n\times n$ symmetric matrix has n distinct real eigenvalues.

3. Problem issues (7 + 9 + 10 + 7 = 33 points)

(1) Let S be the parallelogram determined by the vectors $\begin{pmatrix}1\\3\end{pmatrix}$ and $\begin{pmatrix}5\\1\end{pmatrix}$, and let

$A=\begin{pmatrix}1 & -0.1\\0 & 2\end{pmatrix}$. Compute the area of the image of S under the mapping $x \mapsto Ax$.

(2) Assume the mapping $T: P_2 \to P_2$ defined by

$$T(a_0+a_1t+a_2t^2)=3a_0+(5a_0-2a_1)t+(4a_1+a_2)t^2$$

is linear. Find the matrix representation of T relative to the basis B={1, t, t^2}.

(3) Compute A^k, where $A=\begin{pmatrix}a & 0\\3(a-b) & b\end{pmatrix}$ and k represents an arbitrary positive integer.

(4) Find a least-squares solution of $A\vec{x}=\vec{b}$ for $A=\begin{pmatrix}1 & 0\\1 & 1\\1 & 2\\1 & 3\end{pmatrix}$ and $\vec{b}=\begin{pmatrix}1\\1\\2\\2\end{pmatrix}$.

4. Prove issues (10 + 9 = 19 points)

(1) Prove that a linear transformation T maps R^n onto R^n if and only if T^{-1} exists and also maps R^n onto R^n.

(2) Suppose the solutions of a homogeneous system of five linear equations in six unknowns are all multiples of one nonzero solution. Will the system necessarily be consistent for every possible choice of constants on the right sides of the equations? Explain.

5. Synthesis (8 points)

If $p(t)=c_0+c_1t+\ldots+c_nt^n$, define $p(A)$ to be $p(A)=c_0+c_1A+\ldots+c_nA^n$. Show that if λ is an eigenvalue of A, then one eigenvalue of $p(A)$ is $p(\lambda)$.

《线性代数》期末试题答案 (A/B)

1. Fill in the blank（10*3=30 Pts）

(1) x= $\begin{bmatrix} 11 \\ 2 \end{bmatrix}$

(2) $\det A$ **=6**

(3) a **=8,** b **= -7**

(4) $\hat{y} = \frac{2}{3}\begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix}$

(5) $A = \begin{bmatrix} -2 & 1 \\ 4 & -1 \end{bmatrix}$

(6) A basis for $Row\ A$ **is {（2，-3，6，2，5），（0，0，3，-1，1），（0，0，0，1，3）},**

$\dim Nul\ A$ **is 2，** $rank\ A^T$ **is 3**

(7) $v_1 = \begin{bmatrix} 1+3i \\ 2 \end{bmatrix}$

2. Mark each statement True or False, and descript your reasons. （5*4=20 Pts）

(1) T

(2) T

(3) F

(4) T

(5) F

3. Calculation issues (5*6=30 Pts）

(1)

Solution

a. Determine if the equation $c_1v_1+c_2v_2+c_3v_3=x$ is consistent.

$\begin{bmatrix} 5 & 3 & 1 & -2 \\ 0 & 2 & -1 & 3 \\ 0 & -2 & 1 & 0 \\ 0 & -2 & 1 & -3 \end{bmatrix} \sim \begin{bmatrix} 5 & 3 & 1 & -2 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. **Thus x isn't in W.**

b. Since $y\cdot v_1=0$, $y\cdot v_2=0$, $y\cdot v_3=0$, so y is in $W^{\perp}$.

(2)

Solution

$$AB+I=A^2+B\Rightarrow AB-B=A^2-I\Rightarrow(A-I)\ B=A^2-I\Rightarrow B=(A-I)^{-1}(A^2-I)$$

thus $B=\begin{bmatrix}2&0&1\\0&3&0\\-1&0&2\end{bmatrix}$

(3)

Solution

a. Computer and find $Av_1=-5v_1, Av_2=4v_2, Av_3=4v_3$**, and** $\{v_2,v_3\}$ **must be replace by an orthogonal basis.**

$$\hat{v}_3=(\frac{v_3\cdot v_2}{v_2\cdot v_2})v_2=\frac{1}{2}\begin{bmatrix}1\\1\\0\end{bmatrix},\ v_3-\hat{v}_3=\begin{bmatrix}1\\0\\2\end{bmatrix}-\frac{1}{2}\begin{bmatrix}1\\1\\0\end{bmatrix}=\begin{bmatrix}1/2\\-1/2\\2\end{bmatrix},$$

And normalize the eigenvectors.

Place eigenvectors into $P=\begin{bmatrix}-2/3&1/\sqrt{2}&1/\sqrt{18}\\2/3&1/\sqrt{2}&-1/\sqrt{18}\\1/3&0&4/\sqrt{18}\end{bmatrix}_1$**, and set** $D=\begin{bmatrix}-5&0&0\\0&4&0\\0&0&4\end{bmatrix}_1$

Equation $A=PDP^{-1}$

b. $3x_3^2+8x_1x_2+4x_1x_3-4x_2x_3$**. The quadratic form is indefinite.**

(4)

Solution

$$A^TA=\begin{bmatrix}1&2&3\\-1&5&4\end{bmatrix}\begin{bmatrix}1&-1\\2&5\\3&4\end{bmatrix}=\begin{bmatrix}14&21\\21&42\end{bmatrix},\ A^Tb=\begin{bmatrix}1&2&3\\-1&5&4\end{bmatrix}\begin{bmatrix}-1\\10\\3\end{bmatrix}=\begin{bmatrix}28\\63\end{bmatrix}.$$

Solving $A^TAx=A^Tb$**,** $\begin{bmatrix}14&21&28\\21&42&63\end{bmatrix}$**~…~** $\begin{bmatrix}1&0&-1\\0&1&2\end{bmatrix}$**, so the solution is** $\hat{x}=\begin{bmatrix}-1\\2\end{bmatrix}$**.**

Or $\hat{x}=(A^TA)^{-1}A^Tb=\frac{1}{147}\begin{bmatrix}42&-21\\-21&14\end{bmatrix}\begin{bmatrix}28\\63\end{bmatrix}=\frac{1}{147}\begin{bmatrix}-147\\294\end{bmatrix}=\begin{bmatrix}-1\\2\end{bmatrix}$

And $b-A\hat{x}=\begin{bmatrix}-1\\10\\3\end{bmatrix}-\begin{bmatrix}1&-1\\2&5\\3&4\end{bmatrix}\begin{bmatrix}-1\\2\end{bmatrix}=\begin{bmatrix}2\\2\\-2\end{bmatrix}$**, thus** $\|b-A\hat{x}\|=\sqrt{2^2+2^2+(-2)^2}=\sqrt{12}$

(5)

Solution

A typical element of H can be written as

$$\begin{bmatrix} 3a+7b-c \\ -5b+8c-2d \\ 3d-4e \\ 5b-8c-d+4e \end{bmatrix} = a\begin{bmatrix}3\\0\\0\\0\end{bmatrix} + b\begin{bmatrix}7\\-5\\0\\5\end{bmatrix} + c\begin{bmatrix}-1\\8\\0\\-8\end{bmatrix} + d\begin{bmatrix}0\\-2\\3\\-1\end{bmatrix} + e\begin{bmatrix}0\\0\\-4\\4\end{bmatrix}$$

$\mathbf{u_1}$ $\mathbf{u_2}$ $\mathbf{u_3}$ $\mathbf{u_4}$ $\mathbf{u_5}$

a. H is a vector space because it is the set of all linear combinations of a set of vectors.

b. Row reduce:

$$\begin{bmatrix} 3 & 7 & -1 & 0 & 0 \\ 0 & -5 & 8 & -2 & 0 \\ 0 & 0 & 0 & 3 & -4 \\ 0 & 5 & -8 & -1 & 4 \end{bmatrix} \sim \begin{bmatrix} 3 & 7 & -1 & 0 & 0 \\ 0 & -5 & 8 & -2 & 0 \\ 0 & 0 & 0 & 3 & -4 \\ 0 & 0 & 0 & -3 & 4 \end{bmatrix} \sim \begin{bmatrix} 3 & 7 & -1 & 0 & 0 \\ 0 & -5 & 8 & -2 & 0 \\ 0 & 0 & 0 & 3 & -4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The pivot columns are 1,2, and 4. So { $\mathbf{u_1}$, $\mathbf{u_2}$, $\mathbf{u_4}$} is a basis for H .

4. Prove issues （2*10=20 Pts）

(1)

Proof Since $A^T = (I_n - \frac{2}{\|x\|^2}xx^T)^T = I_n - \frac{2}{\|x\|^2}xx^T = A$, **$A$ is symmetric.**

And because $A^TA = A^2 = (I_n - \frac{2}{\|x\|^2}xx^T)(I_n - \frac{2}{\|x\|^2}xx^T)$

$$= I_n - \frac{4}{\|x\|^2}xx^T + \frac{4}{\|x\|^4}x(x^Tx)x^T$$

$$= I_n - \frac{4}{\|x\|^2}xx^T + \frac{4}{\|x\|^4}x(\|x\|^2)x^T$$

$$= I_n - \frac{4}{\|x\|^2}xx^T + \frac{4}{\|x\|^2}xx^T$$

$$= I_n,$$

thus A is a orthogonal matrix.

Hence A is both orthogonal and symmetric.

(2)

Proof If there is an orthogonal matrix *T* such that $T^{-1}AT = B$, then A is similar to B, thus they have the same characteristic polynomial.

Conversely, if they have the same characteristic polynomial, then they have the same eigenvalues $\lambda_1, \lambda_2, ..., \lambda_n$, thus there is orthogonal matrix T_1 and T_2 such that

$$T_1^{-1}AT_1 = [\lambda_1, \lambda_2, ..., \lambda_n]$$

$$T_2^{-1}BT_2 = [\lambda_1, \lambda_2, ..., \lambda_n]$$

Then

$$T_1^{-1}AT_1 = T_2^{-1}BT_2$$

$$(T_1T_2^{-1})^{-1}AT_1T_2^{-1} = B$$

Let $T = T_1T_2^{-1}$ **, then** T **is an orthogonal matrix such that** $T^{-1}AT = B$

《SE-103 线性代数》期末试题参考答案(A)

1. Fill in the blank（10*3=30 Pts）

(1) x= $\begin{bmatrix}11\\2\end{bmatrix}$

(2) $\det A$ **=6**

(3) a **=8,** b **= -7**

(4) $\hat{y}=\frac{2}{3}\begin{bmatrix}5\\1\\4\end{bmatrix}$

(5) $A=\begin{bmatrix}-2 & 1\\4 & -1\end{bmatrix}$

(6) A basis for $Row\ A$ **is {（2，-3，6，2，5），（0，0，3，-1，1），（0，0，0，1，3）}, (注：原矩阵 A 的前三行线性相关，故不能取 A 的前三行)；** $\dim Nul\ A$ **is 2，** $rank\ A^T$ **is 3**

(7) $v_1=\begin{bmatrix}1+3i\\2\end{bmatrix}$

2. Mark each statement True or False, and descript your reasons. （5*4=20 Pts）

(1) T
(2) T
(3) F
(4) T
(5) F

3. Calculation issues (5*6=30 Pts）

(1)

Solution

a. Determine if the equation $c_1v_1+c_2v_2+c_3v_3=x$ is consistent.

$\begin{bmatrix}5 & 3 & 1 & -2\\0 & 2 & -1 & 3\\0 & -2 & 1 & 0\\0 & -2 & 1 & -3\end{bmatrix}\sim\begin{bmatrix}5 & 3 & 1 & -2\\0 & 2 & -1 & 3\\0 & 0 & 0 & 0\\0 & 0 & 0 & 0\end{bmatrix}$. **Thus x is in W.**

b. Since $y\cdot v_1=0$, $y\cdot v_2=0$, $y\cdot v_3=0$, so y is in $W^{\perp}$.

(2)

Solution

$$AB+I=A^2+B\Rightarrow AB-B=A^2-I\Rightarrow(A-I)\ B=A^2-I\Rightarrow B=(A-I)^{-1}(A^2-I)$$

thus $B=\begin{bmatrix}2&0&1\\0&3&0\\-1&0&2\end{bmatrix}$

(3)

Solution

a. Computer and find $Av_1=-5v_1, Av_2=4v_2, Av_3=4v_3$**, so take**

$\boldsymbol{P=}\begin{bmatrix}\mathbf{-2}&\mathbf{1}&\mathbf{1}\\\mathbf{2}&\mathbf{1}&\mathbf{0}\\\mathbf{1}&\mathbf{0}&\mathbf{2}\end{bmatrix}$ **and** $D=\begin{bmatrix}-5&0&0\\0&4&0\\0&0&4\end{bmatrix}_1$**, we have** $\boldsymbol{A=PDP^{-1}}$

b. $\{v_2,v_3\}$ **must be replace by an orthogonal basis.**

$$\hat{v}_3=(\frac{v_3\cdot v_2}{v_2\cdot v_2})v_2=\frac{1}{2}\begin{bmatrix}1\\1\\0\end{bmatrix},\ v_3-\hat{v}_3=\begin{bmatrix}1\\0\\2\end{bmatrix}-\frac{1}{2}\begin{bmatrix}1\\1\\0\end{bmatrix}=\begin{bmatrix}1/2\\-1/2\\2\end{bmatrix},$$

And normalize the eigenvectors.

Place eigenvectors into $P=\begin{bmatrix}-2/3&1/\sqrt{2}&1/\sqrt{18}\\2/3&1/\sqrt{2}&-1/\sqrt{18}\\1/3&0&4/\sqrt{18}\end{bmatrix}_1$**, and set** $D=\begin{bmatrix}-5&0&0\\0&4&0\\0&0&4\end{bmatrix}_1$

Equation $\boldsymbol{A=QDQ^T}$**.**

(4)

Solution

$$A^TA=\begin{bmatrix}1&2&3\\-1&5&4\end{bmatrix}\begin{bmatrix}1&-1\\2&5\\3&4\end{bmatrix}=\begin{bmatrix}14&21\\21&42\end{bmatrix},\ A^Tb=\begin{bmatrix}1&2&3\\-1&5&4\end{bmatrix}\begin{bmatrix}-1\\10\\3\end{bmatrix}=\begin{bmatrix}28\\63\end{bmatrix}.$$

Solving $A^TAx=A^Tb$**,** $\begin{bmatrix}14&21&28\\21&42&63\end{bmatrix}$**~…~** $\begin{bmatrix}1&0&-1\\0&1&2\end{bmatrix}$**, so the solution is** $\hat{x}=\begin{bmatrix}-1\\2\end{bmatrix}$**.**

Or $\hat{x}=(A^TA)^{-1}A^Tb=\frac{1}{147}\begin{bmatrix}42&-21\\-21&14\end{bmatrix}\begin{bmatrix}28\\63\end{bmatrix}=\frac{1}{147}\begin{bmatrix}-147\\294\end{bmatrix}=\begin{bmatrix}-1\\2\end{bmatrix}$

And $b-A\hat{x}=\begin{bmatrix}-1\\10\\3\end{bmatrix}-\begin{bmatrix}1&-1\\2&5\\3&4\end{bmatrix}\begin{bmatrix}-1\\2\end{bmatrix}=\begin{bmatrix}2\\2\\-2\end{bmatrix}$**, thus** $\|b-A\hat{x}\|=\sqrt{2^2+2^2+(-2)^2}=\sqrt{12}$

(5)

Solution

A typical element of H can be written as

$$\begin{bmatrix} 3a+7b-c \\ -5b+8c-2d \\ 3d-4e \\ 5b-8c-d+4e \end{bmatrix} = a\begin{bmatrix}3\\0\\0\\0\end{bmatrix} + b\begin{bmatrix}7\\-5\\0\\5\end{bmatrix} + c\begin{bmatrix}-1\\8\\0\\-8\end{bmatrix} + d\begin{bmatrix}0\\-2\\3\\-1\end{bmatrix} + e\begin{bmatrix}0\\0\\-4\\4\end{bmatrix}$$

$\mathbf{u_1}$ $\mathbf{u_2}$ $\mathbf{u_3}$ $\mathbf{u_4}$ $\mathbf{u_5}$

a. H is a vector space because it is the set of all linear combinations of a set of vectors.

b. Row reduce:

$$\begin{bmatrix} 3 & 7 & -1 & 0 & 0 \\ 0 & -5 & 8 & -2 & 0 \\ 0 & 0 & 0 & 3 & -4 \\ 0 & 5 & -8 & -1 & 4 \end{bmatrix} \sim \begin{bmatrix} 3 & 7 & -1 & 0 & 0 \\ 0 & -5 & 8 & -2 & 0 \\ 0 & 0 & 0 & 3 & -4 \\ 0 & 0 & 0 & -3 & 4 \end{bmatrix} \sim \begin{bmatrix} 3 & 7 & -1 & 0 & 0 \\ 0 & -5 & 8 & -2 & 0 \\ 0 & 0 & 0 & 3 & -4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The pivot columns are 1,2, and 4. So { $\mathbf{u_1}$, $\mathbf{u_2}$, $\mathbf{u_4}$} is a basis for H .

4. Prove issues （2*10=20 Pts）

(1)

Proof Since $A^T = (I_n - \frac{2}{\|x\|^2}xx^T)^T = I_n - \frac{2}{\|x\|^2}xx^T = A$, **$A$ is symmetric.**

And because $A^TA = A^2 = (I_n - \frac{2}{\|x\|^2}xx^T)(I_n - \frac{2}{\|x\|^2}xx^T)$

$$= I_n - \frac{4}{\|x\|^2}xx^T + \frac{4}{\|x\|^4}x(x^Tx)x^T$$

$$= I_n - \frac{4}{\|x\|^2}xx^T + \frac{4}{\|x\|^4}x(\|x\|^2)x^T$$

$$= I_n - \frac{4}{\|x\|^2}xx^T + \frac{4}{\|x\|^2}xx^T$$

$$= I_n,$$

thus A is a orthogonal matrix.

Hence A is both orthogonal and symmetric.

(2)

Proof Ax=(I-2xx^T)x=x-2x(x^{T}x)=-x, so –1 is an eigenvalue, the eigenspace is span{x}.

Take a nonzero vector v in (span{ x })$^\perp$, Av=(I-2xx^T)v=v-2x(x^{T}v)=v, so 1 is the other

eigenvalue, its eigenspace is (span{ x })$^{\perp}$.

《SE-103 线性代数》期末试题参考答案(B)

1.（每小题 4 分）

(1)10; **(2)** $\begin{pmatrix}-2\\2\\-3\end{pmatrix}, \begin{pmatrix}4\\-6\\8\end{pmatrix}$; **(3)** $-\lambda^3+4\lambda^2-9\lambda-6$; **(4)** $\begin{pmatrix}-2\\1\end{pmatrix}$;

(5) $\begin{pmatrix}x_1 & x_2 & x_3\end{pmatrix}\begin{pmatrix}5 & -3 & 0\\-3 & 3 & 4\\0 & 4 & 2\end{pmatrix}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}$

2. （判断正确 2 分，说理充分 2 分）

(1)错误，当 m=n 时才是一对一的映射。

(2)错误，因为矩阵的行阶梯形式并不唯一。

(3)错误，因为例如令 A 和 B 为两个不同的 2 阶置换矩阵，则（A+B）不可逆。

(4)正确，因为两者有相同的特征多项式。

(5)错误，例如单位矩阵是对称的，但是没有不同的特征值。

3.

(1)（过程正确 5 分，计算正确 2 分）

S 的面积是 $\left|\det\begin{pmatrix}1 & 5\\3 & 1\end{pmatrix}\right|=14$**, det A=2, 所以根据第 3 章定理 10 可知，线性变换后 S 的面积是 2×14=28.**

(2)（各个基 3 分，其中过程正确 2 分，结果正确 1 分）

根据题意可知，有 T(1)=3+5t, T(t)=-2t+4t^2, T(t^2)=t^2, 所以它对应以 B 为底的变换矩阵为

$$\begin{pmatrix}3 & 0 & 0\\5 & -2 & 0\\0 & 4 & 1\end{pmatrix}$$

(3)（特征值求解 6 分，其中每个特征值 1 分，对应特征向量 2 分；矩阵对角化分解 2 分；最后计算结果 2 分，没有化简则只给 1 分）

根据题意可知，由于原矩阵是下三角阵，所以其特征值为主对角线上的元素，即 a 和 b。

代入后分别计算得其对应特征向量为 $v_a=\begin{pmatrix}1\\3\end{pmatrix}, v_b=\begin{pmatrix}0\\1\end{pmatrix}$**,所以原矩阵可以对角化为**

$$\begin{pmatrix} a & 0 \\ 3(a-b) & b \end{pmatrix} = PDP^{-1} = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix}$$

所以有

$$\begin{pmatrix} a & 0 \\ 3(a-b) & b \end{pmatrix}^k = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}\begin{pmatrix} a^k & 0 \\ 0 & b^k \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} a^k & 0 \\ 3(a^k-b^k) & b^k \end{pmatrix}$$

(4) (计算过程正确 5 分，计算结果正确 2 分)

根据最小二乘法可知，方程 $A\vec{x} = \vec{b}$ 的最小二乘解满足方程

$$A^T A\vec{x} = A^T\vec{b}$$

又因为

$$A^T A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix} \qquad A^T\vec{b} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} \quad \\ 1 \end{pmatrix}$$

解得 $x_1 = 0.9$， $x_2 = 0.4$，即为所求方程的最小二乘解。

4. (1)（充分性与必要性证明各占 5 分）

证：记线性变换 T 的标准矩阵为 A，则由于 T 是 R^n 到 R^n 的满射，所以 A 可逆是其充分必要条件。根据第二章定理 9 可知，线性变换 T 可逆也是 A 可逆充分必要条件。

(2)（判断 3 分，理由说明 6 分，课本 270 页习题 19）

解：一定有解。因为记原线性方程组的系数矩阵为 A，则由题意可知 A 是一个 5×6 的矩阵且原线性方程组的非平凡解中只有一个变量非零；所以 dim Nul A = 1。因此根据秩定理可知 dim Col A = rank A = 6 − 1 = 5，而 R^5 中只有它本身作为子集时维数才是 5。这表示 Col A 就是 R^5，即 A 为系数矩阵的任意一个线性方程组都是相容的。

5.证：

记矩阵 A 的特征值 λ 对应的特征向量是 $\vec{v}$，则根据定义可知 $A\vec{v} = \lambda\vec{v}$，所以有

$$p(A)\vec{v} = (c_0 + c_1A + \cdots + c_nA^n)\vec{v} = (c_0 + c_1\lambda + \cdots + c_n\lambda^n)\vec{v} = p(\lambda)\vec{v}$$

这表示 $p(\lambda)$ 就是 $p(A)$ 的一个特征值。

（8 分）

中山大学软件学院 **2010** 级软件工程专业(2010学年秋季学期)

《线性代数》期末试题(A/B卷)

(考试形式：闭　卷　　考试时间：2小时)

警示

《中山大学授予学士学位工作细则》第六条

考试作弊不授予学士学位

方向：_____　　姓名：_____　　学号：_____

出卷：_____　　复核：_____

1. Fill in the blank（10*3=30 Pts）

(1) If the vector x determined by the coordinate vector $[x]_B = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$ **and the basis** $B = \left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \end{bmatrix} \right\}$**, then x=_____.**

(2) Let A **be a 4×4 matrix, and suppose the eigenvalues of** A **are 3，1，1，and 2, then** $\det A$ **=_____**

(3) The matrix $A = \frac{1}{9}\begin{bmatrix} -1 & 4 & a \\ a & 4 & -1 \\ 4 & b & 4 \end{bmatrix}$ **is an orthogonal matrix, then** a **=_____,** b **=_____.**

(4) If $v_1 = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}, v_2 = \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix}, y = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$**, and** $W = Span\{v_1, v_2\}$**, then the closest point in** W **to** y **is _____.**

(5) If the transformation $x \mapsto Ax$ **maps** $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ **and** $\begin{bmatrix} 2 \\ 7 \end{bmatrix}$ **into** $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ **and** $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$**,respectively, then the matrix** A **=_____.**

(6) Assume the matrix A **is row equivalent to** B**.**

$$A=\begin{bmatrix} 2 & -3 & 6 & 2 & 5 \\ -2 & 3 & -3 & -3 & -4 \\ 4 & -6 & 9 & 5 & 9 \\ -2 & 3 & 3 & -4 & 1 \end{bmatrix}, \quad B=\begin{bmatrix} 2 & -3 & 6 & 2 & 5 \\ 0 & 0 & 3 & -1 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

A basis for $Row\ A$ is _____, and $\dim Nul\ A$ is_____, and $rank\ A^T$ is _____.

(7) $A=\begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix}$ act on C^n, the eigenvalues of A is $\lambda = 2 \pm 3i$, and a basis for the eigenspace corresponding to $\lambda_1 = 2-3i$ is _____.

2. Mark each statement True or False, and descript your reasons. （5*4=20 Pts）

(1) Let $T: R^n \mapsto R^n$ be a linear transformation defined by $T(x)=Ax$. Then T is onto if and only if $\det(A) \neq 0$.

(2) If A is an *n*×*n* matrix, then $rank\ A < n$ if and only if some eigenvalue of A is zero.

(3) The set of all solution to the linear system $Ax=b$, where A is *m*×*n* and $b \neq 0$, is a subspace of R^n.

(4) Every set of five orthonormal vectors is a basis for R^5.

(5) The linear transformation $T: P_2 \mapsto P_2$ defined by $T(at^2+bt+c)=2at+b$ is one-to-one.

3. Calculation issues (5*6=30 Pts）

(1) Let $v_1=\begin{bmatrix} 5 \\ 0 \\ 0 \\ 0 \end{bmatrix}, v_2=\begin{bmatrix} 3 \\ 2 \\ -2 \\ -2 \end{bmatrix}, v_3=\begin{bmatrix} 2 \\ -1 \\ 1 \\ 1 \end{bmatrix}, x=\begin{bmatrix} -2 \\ 3 \\ 0 \\ -3 \end{bmatrix}, y=\begin{bmatrix} 0 \\ 3 \\ 6 \\ -3 \end{bmatrix}$, and $W=Span\{v_1, v_2, v_3\}$

a. Determine whether x is in W

b. Determine whether y is in the orthogonal of W.

(2) If A and B are 3×3 matrices, I is the identity matrix, and $AB+I=A^2+B$, where $A=\begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ -1 & 0 & 1 \end{bmatrix}$. Find B.

(3) Let $A=\begin{bmatrix}0 & 4 & 2\\ 4 & 0 & -2\\ 2 & -2 & 3\end{bmatrix}$, $v_1=\begin{bmatrix}-2\\ 2\\ 1\end{bmatrix}$, $v_2=\begin{bmatrix}1\\ 1\\ 0\end{bmatrix}$, $v_3=\begin{bmatrix}1\\ 0\\ 2\end{bmatrix}$. **The** v_1, v_2, **and** v_3 **are eigenvectors for** A.

a. Find matrices P **and** D **that orthogonally diagonalize** A, **and write the equation that relates** A **to** P **and** D.

b. Write the quadratic form associated with A **using variables** x_1, x_2, **and** x_3. **And classify the quadratic form as positive (or negative) definite, or indefinite.**

(4) Let $A=\begin{bmatrix}1 & -1\\ 2 & 5\\ 3 & 4\end{bmatrix}$, $b=\begin{bmatrix}-1\\ 10\\ 3\end{bmatrix}$. **Find the least-squares solution of** $Ax=b$, **and determine the least-squares error in the least-squares solution of** $Ax=b$.

(5) Let $H=\left\{\begin{bmatrix}3a+7b-c\\ -5b+8c-2d\\ 3d-4e\\ 5b-8c-d+4e\end{bmatrix}: a,b,c,d,e \text{ any real numbers}\right\}$,

a. Show H **is a subspace of** R^4.

b. Find a basis for H .

4. Prove issues （2*10=20 Pts）

(1) Show that if x is a nonzero vector in R^n, **then the** $n\times n$ **matrix**

$$A=I_n-\frac{2}{\|x\|^2}xx^T$$

is both orthogonal and symmetric.

(2) Suppose A **and** B **are both real symmetric matrix, then they have the same characteristic polynomial if and only if there is an orthogonal matrix** T **such that** $T^{-1}AT=B$.

中山大学软件学院 **2010** 级软件工程专业(2010学年秋季学期)

《SE-103 线性代数》期末试题(A卷)

(考试形式：闭 卷 考试时间：2小时)

警示

《中山大学授予学士学位工作细则》第六条

考试作弊不授予学士学位

方向：_____ 姓名：_____ 学号：_____

出卷：伍丽华 复核：何伟弘

1. Fill in the blank（10*3=30 Pts）

(1) If the vector *x* determined by the coordinate vector $[x]_B = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$ **and the basis** $B = \left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \end{bmatrix} \right\}$**, then *x*=_____.**

(2) Let A **be a 4×4 matrix, and suppose the eigenvalues of** A **are 3，1，1，and 2, then** $\det A$ **=_____**

(3) The matrix $A = \frac{1}{9}\begin{bmatrix} -1 & 4 & a \\ a & 4 & -1 \\ 4 & b & 4 \end{bmatrix}$ **is an orthogonal matrix, then** a **=_____,** b **=_____.**

(4) If $v_1 = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}, v_2 = \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix}, y = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$**, and** $W = Span\{v_1, v_2\}$**, then the closest point in** W **to** y **is _____.**

(5) If the transformation $x \mapsto Ax$ **maps** $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ **and** $\begin{bmatrix} 2 \\ 7 \end{bmatrix}$ **into** $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ **and** $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$**,respectively, then the matrix** A **=_____.**

(6) Assume the matrix A **is row equivalent to** B**.**

$$A=\begin{bmatrix}2&-3&6&2&5\\-2&3&-3&-3&-4\\4&-6&9&5&9\\-2&3&3&-4&1\end{bmatrix},\quad B=\begin{bmatrix}2&-3&6&2&5\\0&0&3&-1&1\\0&0&0&1&3\\0&0&0&0&0\end{bmatrix}$$

A basis for $Row\ A$ **is** _____, **and** $\dim Nul\ A$ **is**_____, **and** $rank\ A^T$ **is** _____.

(7) $A=\begin{bmatrix}1&5\\-2&3\end{bmatrix}$ **act on** C^n, **the eigenvalues of** A **is** $\lambda=2\pm3i$, **and a basis for the eigenspace corresponding to** $\lambda_1=2-3i$ **is** _____.

2. Mark each statement True or False, and descript your reasons. （5*4=20 Pts）

(1) Let $T: R^n \to R^n$ **be a linear transformation defined by** $T(x)=Ax$. **Then** T **is onto if and only if** $\det(A)\neq 0$.

(2) If A **is an *n*×*n* matrix, then** $rank\ A<n$ **if and only if zero is an eigenvalue of** A.

(3) The set of all solutions to the linear system $Ax=b$, **where** A **is *m*×*n* and** $b\neq 0$, **is a subspace of** R^n.

(4) Every set of five orthonormal vectors is a basis for R^5.

(5) The linear transformation $T: P_2 \to P_2$ **defined by** $T(at^2+bt+c)=2at+b$ **is one-to-one.**

3. Calculation issues (5*6=30 Pts）

(1) Let $v_1=\begin{bmatrix}5\\0\\0\\0\end{bmatrix}, v_2=\begin{bmatrix}3\\2\\-2\\-2\end{bmatrix}, v_3=\begin{bmatrix}2\\-1\\1\\1\end{bmatrix}, x=\begin{bmatrix}-2\\3\\0\\-3\end{bmatrix}, y=\begin{bmatrix}0\\3\\6\\-3\end{bmatrix}$, **and** $W=Span\{v_1,v_2,v_3\}$

a. Determine whether *x* is in W

b. Determine whether *y* is in the orthogonal complement of W.

(2) If A **and** B **are 3×3 matrices,** I **is the identity matrix, and** $AB+I=A^2+B$, **where** $A=\begin{bmatrix}1&0&1\\0&2&0\\-1&0&1\end{bmatrix}$. **Find** B.

(3) Let $A=\begin{bmatrix}0 & 4 & 2\\ 4 & 0 & -2\\ 2 & -2 & 3\end{bmatrix}$, $v_1=\begin{bmatrix}-2\\ 2\\ 1\end{bmatrix}$, $v_2=\begin{bmatrix}1\\ 1\\ 0\end{bmatrix}$, $v_3=\begin{bmatrix}1\\ 0\\ 2\end{bmatrix}$, **and** v_1, v_2, **and** v_3 **are eigenvectors for** A.

a. Diagonalize *A*. (Find an invertible matix *P* and a diagonal matrix *D* such that $A=PDP^{-1}$)

b. Use orthogonal matrix to Diagonalize *A*. (Find an orthogonal matrix Q such that $A=QDQ^{T}$)

(4) Let $A=\begin{bmatrix}1 & -1\\ 2 & 5\\ 3 & 4\end{bmatrix}$, $b=\begin{bmatrix}-1\\ 10\\ 3\end{bmatrix}$. **Find the least-squares solution of** $Ax=b$, **and determine the least-squares error in the least-squares solution of** $Ax=b$.

(5) Let $H=\left\{\begin{bmatrix}3a+7b-c\\ -5b+8c-2d\\ 3d-4e\\ 5b-8c-d+4e\end{bmatrix}: a,b,c,d,e \text{ any real numbers}\right\}$,

a. Show that H **is a subspace of** R^4.

b. Find a basis for H .

4. Prove issues （2*10=20 Pts）

(1) Let *x* be a unit vector in R^n, **prove that the** $n\times n$ **matrix** $A=I_n-2xx^T$ **is both orthogonal and symmetric.**

(2) Let *x* be a unit vector in R^3, prove that $A=I_3-2xx^T$ **has eigenvalues** $\lambda=1$ **and** $\lambda=-1$, **furthermore, descript the corresponding eigenspaces. (Hint: the transformatiom** $v\mapsto Av$ **reflects points across the plane (span{ *x* })$^{\perp}$)**

中山大学软件学院 **2010** 级软件工程专业(2010学年秋季学期)

《SE-103 线性代数》期末试题(B卷)

(考试形式：闭 卷 考试时间：2小时)

警示

《中山大学授予学士学位工作细则》第六条

考试作弊不授予学士学位

方向：_____ 姓名：_____ 学号：_____

出卷：高成英 复核：何伟弘

1. Fill the blank (5 titles * 4 points/title = 20 points)

(1) The determinant of $\begin{vmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{vmatrix}$ is_____.

(2) Assume that $A = \begin{pmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{pmatrix}$ is row equivalent to $\begin{pmatrix} 1 & 0 & 6 & 5 \\ 0 & 2 & 5 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$. A basis for Col A is _____.

(3) The characteristic polynomial of $\begin{pmatrix} 1 & 0 & -1 \\ 2 & 3 & -1 \\ 0 & 6 & 0 \end{pmatrix}$ is _____.

(4) The orthogonal projection of $\begin{pmatrix} 1 \\ 7 \end{pmatrix}$ onto the line through $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$ and the origin is _____.

(5) For $\vec{x}$ in $\mathbf{R}^3$, let $Q(\vec{x}) = 5x_1^2 + 3x_2^2 + 2x_3^2 - 6x_1x_2 + 8x_2x_3$. Then the matrix of the quadratic form is _____.

2. Mark each statement True or False, and descript your reasons (5titles*4pts/title=20pts)

(1) If A is $m \times n$ and rank $A=m$, then the linear transformation $x \mapsto Ax$ is one-to-one.

(2) If matrices A and B are row equivalent, they have the same echelon form.

(3) If matrices A and B are both $n \times n$ invertible matrices, then $(A+B)$ is also invertible.

(4) The matrices A and A^T have the same eigenvalues, counting multiplicities.

(5) An $n \times n$ symmetric matrix has n distinct real eigenvalues.

3. Problem issues (7 + 9 + 10 + 7 = 33 points)

(1) Let S be the parallelogram determined by the vectors $\begin{pmatrix}1\\3\end{pmatrix}$ **and** $\begin{pmatrix}5\\1\end{pmatrix}$**, and let**

$A=\begin{pmatrix}1 & -0.1\\0 & 2\end{pmatrix}$. **Compute the area of the image of S under the mapping** $x \mapsto Ax$.

(2) Assume the mapping $T: P_2 \to P_2$ **defined by**

$$T(a_0+a_1t+a_2t^2)=3a_0+(5a_0-2a_1)t+(4a_1+a_2)t^2$$

is linear. Find the matrix representation of T **relative to the basis B={1,** t**,** t^2**}.**

(3) Compute A^k**, where** $A=\begin{pmatrix}a & 0\\3(a-b) & b\end{pmatrix}$ **and** k **represents an arbitrary positive integer.**

(4) Find a least-squares solution of $A\vec{x}=\vec{b}$ **for** $A=\begin{pmatrix}1 & 0\\1 & 1\\1 & 2\\1 & 3\end{pmatrix}$ **and** $\vec{b}=\begin{pmatrix}1\\1\\2\\2\end{pmatrix}$.

4. Prove issues (10 + 9 = 19 points)

(1) Prove that a linear transformation T **maps** R^n **onto** R^n **if and only if** T^{-1} **exists and also maps** R^n **onto** R^n**.**

(2) Suppose the solutions of a homogeneous system of five linear equations in six unknowns are all multiples of one nonzero solution. Will the system necessarily be consistent for every possible choice of constants on the right sides of the equations? Explain.

5. Synthesis (8 points)

If $p(t)=c_0+c_1t+\ldots+c_nt^n$**, define** $p(A)$ **to be** $p(A)=c_0+c_1A+\ldots+c_nA^n$**. Show that if** λ **is an eigenvalue of** A**, then one eigenvalue of** $p(A)$ **is** $p(\lambda)$**.**

1.（1）10；　（2）$\begin{pmatrix}-2\\2\\-3\end{pmatrix},\begin{pmatrix}4\\-6\\8\end{pmatrix}$；　（3）$-\lambda^3+4\lambda^2-9\lambda-6$；　(4) $\begin{pmatrix}-2\\1\end{pmatrix}$

2.（判断正确 2 分，理由正确 2 分）
(1)错误，因为映射后的向量是 n 维的，所以只有当矩阵的秩是 n 时才是一对一的映射。
(2)错误，因为矩阵的行阶梯形式并不唯一。
(3)错误，因为例如令 A 和 B 为两个不同的 2 阶置换矩阵，则（A+B）不可逆。
(4)正确，因为两者有相同的特征多项式。
(5)错误，例如单位矩阵是对称的，但是没有不同的特征值。

3.
(1)（过程正确 5 分，计算正确 2 分）

S 的面积是$\left|\det\begin{pmatrix}1&5\\3&1\end{pmatrix}\right|=14$，又因为$\det A=2$，所以根据第 3 章定理 10 可知，线性变换后 S 的面积是 2×14=28。.

(2) （各个基 3 分，其中过程正确 2 分，结果正确 1 分）

根据题意可知，有$T(1)=3+5t$，$T(t)=-2t+4t^2$ 和$T(t^2)=t^2$，所以它对应以 B 为底的变换矩阵为

$$\begin{pmatrix}3&0&0\\5&-2&0\\0&4&1\end{pmatrix}$$

(3)（特征值求解 6 分，其中每个特征值 1 分，对应特征向量 2 分；矩阵对角化分解 2 分；最后计算结果 2 分，没有化简则只给 1 分）
根据题意可知， 由于原矩阵是下三角阵，所以其特征值为主对角线上的元素，即 a 和 b。

代入后分别计算得其对应特征向量为$\vec{v}_a=\begin{pmatrix}1\\3\end{pmatrix}$和$\vec{v}_b=\begin{pmatrix}0\\1\end{pmatrix}$，所以原矩阵可以对角化为

$$\begin{pmatrix}a&0\\3(a-b)&b\end{pmatrix}=PDP^{-1}=\begin{pmatrix}1&0\\3&1\end{pmatrix}\begin{pmatrix}a&0\\0&b\end{pmatrix}\begin{pmatrix}1&0\\-3&1\end{pmatrix}$$

所以有

$$\begin{pmatrix}a&0\\3(a-b)&b\end{pmatrix}^k=\begin{pmatrix}1&0\\3&1\end{pmatrix}\begin{pmatrix}a^k&0\\0&b^k\end{pmatrix}\begin{pmatrix}1&0\\-3&1\end{pmatrix}=\begin{pmatrix}a^k&0\\3(a^k-b^k)&b^k\end{pmatrix}$$

(4)（列向量 Gram-Schmidt 正交化正确 6 分，其中每个正交基过程正确 1 分，计算正确 1 分；R 矩阵计算准确 2 分，过程正确 3 分）

根据题目可知，原始矩阵各列由 Gram-Schmidt 正交化得

$$\vec{v}_1=\begin{pmatrix}1/\sqrt{5}\\-1/\sqrt{5}\\-1/\sqrt{5}\\1/\sqrt{5}\\1/\sqrt{5}\end{pmatrix}\quad \vec{v}_2=\begin{pmatrix}1/2\\0\\1/2\\-1/2\\1/2\end{pmatrix}\quad \vec{v}_3=\begin{pmatrix}1/2\\0\\1/2\\1/2\\-1/2\end{pmatrix}$$

所以上三角阵 R 为

$$R=\begin{pmatrix}1/\sqrt{5} & -1/\sqrt{5} & -1/\sqrt{5} & 1/\sqrt{5} & 1/\sqrt{5}\\ 1/2 & 0 & 1/2 & -1/2 & 1/2\\ 1/2 & 0 & 1/2 & 1/2 & -1/2\end{pmatrix}\begin{pmatrix}1 & 2 & 5\\ -1 & 1 & -4\\ -1 & 4 & -3\\ 1 & -4 & 7\\ 1 & 2 & 1\end{pmatrix}=\begin{pmatrix}\sqrt{5} & -\sqrt{5} & 4\sqrt{5}\\ 0 & 6 & -2\\ 0 & 0 & 4\end{pmatrix}$$

4.

(1)（充分性与必要性证明各占 5 分）

证：记线性变换 T 的标准矩阵为 A，则由于 T 是 R^n 到 R^n 的满射，所以 A 可逆是其充分必要条件。根据第二章定理 9 可知，线性变换 T 可逆也是 A 可逆充分必要条件。

(2)（判断 3 分，理由说明 6 分）

解：一定有解。因为记原线性方程组的系数矩阵为 A，则由题意可知 A 是一个 5×6 的矩阵且原线性方程组的非平凡解中只有一个变量非零；所以 dim Nul A = 1。因此根据秩定理可知 dim Col A = rank A = 6 – 1 = 5，而 R^5 中只有它本身作为子集时维数才是 5。这表示 Col A 就是 R^5，即 A 为系数矩阵的任意一个线性方程组都是相容的。

5.证：

记矩阵 A 的特征值 λ 对应的特征向量是 $\vec{v}$，则根据定义可知 $A\vec{v}=\lambda\vec{v}$，所以有

$$p(A)\vec{v}=(c_0+c_1A+\cdots+c_nA^n)\vec{v}=(c_0+c_1\lambda+\cdots+c_n\lambda^n)\vec{v}=p(\lambda)\vec{v}$$

这表示 $p(\lambda)$ 就是 $p(A)$ 的一个特征值。

软件学院 2010 级软件工程专业(2011-1)

《线性代数》期末试题试卷

(考试形式： 闭 卷 考试时间：120 分钟)

警示

《中山大学授予学士学位工作细则》第六条

考试作弊不授予学士学位

方向：_____ 姓名：_____ 学号：_____成绩：_____

1. Fill the blank (4 titles * 4 points/title = 16 points)

(1) The determinant of $\begin{vmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{vmatrix}$ is _____.

(2) Assume that $A=\begin{pmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{pmatrix}$ is row equivalent to $\begin{pmatrix} 1 & 0 & 6 & 5 \\ 0 & 2 & 5 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$. Bases for Col A are _____.

(3) The characteristic polynomial of $\begin{pmatrix} 1 & 0 & -1 \\ 2 & 3 & -1 \\ 0 & 6 & 0 \end{pmatrix}$ is _____.

(4) The orthogonal projection of $\begin{pmatrix} 1 \\ 7 \end{pmatrix}$ onto the line through $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$ and the origin is _____.

2. Mark each statement True or False, and descript your reasons (5titles * 4 points/title = 20 points)

(1) If A is m × n and rank A = m, then the linear transformation $\vec{x} \mapsto A\vec{x}$ is one-to-one.

(2) If matrices A and B are row equivalent, they have the same echelon form.

(3) If matrices A and B are both n × n invertible matrices, then $(A+B)$ is also invertible.

(4) The matrices A and A^{T} have the same eigenvalues, counting multiplicities.

(5) An n × n symmetric matrix has n distinct real eigenvalues.

3. Problem issues (7 + 9 + 10 + 11 = 37 points)

(1) Let S be the parallelogram determined by the vectors $\begin{pmatrix}1\\3\end{pmatrix}$ and $\begin{pmatrix}5\\1\end{pmatrix}$, and let $A=\begin{pmatrix}1 & -0.1\\0 & 2\end{pmatrix}$. Computer the area of the image of S under the mapping $\vec{x}\mapsto A\vec{x}$ (7 points)

(2) Assume the mapping $T:P_2\to P_2$ defined by

$$T(a_0+a_1t+a_2t^2)=3a_0+(5a_0-2a_1)t+(4a_1+a_2)t^2$$

is linear. Find the matrix representation of T relative to the basis $B=\{1,t,t^2\}$ (9 points).

(3) Computer A^{k}, where $A=\begin{pmatrix}a & 0\\3(a-b) & b\end{pmatrix}$ and k represents an arbitrary positive integer (10 points).

(4) Find a QR factorization of the matrix $\begin{pmatrix}1 & 2 & 5\\-1 & 1 & -4\\-1 & 4 & -3\\1 & -4 & 7\\1 & 2 & 1\end{pmatrix}$ (11 points).

4. Prove issues (10 + 9 = 19 points)

(1) Prove that a linear transformation T maps R^{n} onto R^{n} if and only if T^{-1} exists and also maps R^{n} onto R^{n} (10 points).

(2) Suppose the solutions of a homogeneous system of five linear equations in six unknowns are all multiples of one nonzero solution. Will the system necessarily be consistent for every possible choice of constants on the right sides of the equations? Explain.(9 points)

5. Synthesis (8 points)

If $p(t) = c_0 + c_1 t + \cdots + c_n t^n$, define $p(A)$ to be $p(A) = c_0 + c_1 A + \cdots + c_n A^n$. Show that if λ is an eigenvalue of A, then one eigenvalue of $p(A)$ is $p(\lambda)$ (8 points).

中山大学软件学院 **2011** 级软件工程专业(2011学年秋季学期)

《SE-103+线性代数》期末试题(B卷)

(考试形式：闭　卷　　考试时间：2小时)

警示

《中山大学授予学士学位工作细则》第六条

考试作弊不授予学士学位

方向：_____　　姓名：_____　　学号：_____

出卷：伍丽华　　复核：高成英

1. Fill in the blank（5×4=20 Pts）

(1) If T **is the linear transformation from** P_2 **to** P_2 **whose matrix relative to** $B=\{1, t, t^2\}$ **is**

$[T]_B=\begin{bmatrix}2 & 1 & 0\\ 0 & 3 & -1\\ 1 & -2 & 4\end{bmatrix}$**, then** $T(a_0+a_1t+a_2t^2)$=_____.

(2) If the row space of a 4×7 matrix A **is 4-dimentional, then the dimension of the null space of** A **is** _____. **Is** $\text{Col } A=R^4$**?** _____ **(Yes or No).**

(3) Let $v_1=\begin{bmatrix}2\\2\\0\end{bmatrix}$, $v_2=\begin{bmatrix}2\\2\\2\end{bmatrix}$, **and** $v_3=\begin{bmatrix}0\\2\\2\end{bmatrix}$ **be eigenvectors of a 3×3 matrix** A**, with corresponding eigenvalues 3, 2, and 1. Compute** A **.** A =_____.

(4) Determine the value(s) of a **such that the system** $\begin{bmatrix}1 & 2 & 1\\ 2 & 3 & a+2\\ 1 & a & -2\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}=\begin{bmatrix}1\\3\\0\end{bmatrix}$ **is inconsistent.** a =_____.

(5) For x **in** R^3**, Let** $Q(x)=3x_1^2+5x_2^2-x_1x_2+8x_2x_3$**, this quadratic form as** x^TAx **is** _____.

2. Make each statement True or False, and descript your reasons.（5×4=20 Pts）

(1) Whenever a system has free variables, the solution set contains many solutions.

(2) If $v_1, v_2, \cdots, v_k$ **are vectors in a vector space** V **and**

$\mathrm{Span}\{v_1, v_2, \cdots, v_k\} = \mathrm{Span}\{v_1, v_2, \cdots, v_{k-1}\}$**, then** $v_1, v_2, \cdots, v_k$ **are linearly dependent.**

(3) Let $T: R^n \to R^n$ **be a linear transformation. If** A **is the standard matrix representation of** T**, then an *n*×*n* matrix** B **will also be a matrix representation of** T **if and only if** B **is similar to** A**.**

(4) If A **is an *n*×*n* matrix, then** A **and** A^T **have the same eigenvectors.**

(5) If A **is symmetric and det(**A**)>0, then** A **is positive definite.**

3. Calculation (5×8=40 Pts）

(1) let $A = [b_1 \;\; b_2 \;\; b_3]$ **and** $B = [b_1 + b_2 + b_3 \;\; b_1 + 2b_2 + 4b_3 \;\; b_1 + 3b_2 + 9b_3]$**, where** b_1, b_2 **and** b_3 **are vectors in** R^3**. Suppose** $\det A = 1$**, find** $\det B$**.**

(2) Computer A^6**, where** $A = \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix}$**.**

(3) Let $H = \left\{ \begin{bmatrix} 3a + 7b - c \\ -5b + 8c - 2d \\ 3d - 4e \\ 5b - 8c - d + 4e \end{bmatrix} a, b, c, d, e \text{ any real numbers} \right\}$**,**

a. Show that H **is a subspace of** R^4

b. Find a basis for H**.**

(4) Let $A = \begin{bmatrix} 1 & 5 \\ 3 & 1 \\ -2 & 4 \end{bmatrix}$**,** $b = \begin{bmatrix} 4 \\ -2 \\ -3 \end{bmatrix}$**.**

a. Find the orthogonal projection of b **onto Col** A**.**

b. Find a least-squares solution of $Ax = b$**.**

c. Determine the associated least-squares error.

(5) Let W =Span$\{x_1, x_2\}$, where $x_1 = \begin{bmatrix} 2 \\ -5 \\ 1 \end{bmatrix}$, and $x_2 = \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix}$, Construct an orthonormal basis for W.

4. Prove issues（2×6=12 Pts）

(1) Let W be a subspace of R^n such that $\dim W = p$, and let $S = \{w_1, w_2, \cdots, w_p\}$ be an orthonomal basis for W. Define $T : R^n \to W$ by

$$T(v) = (v \cdot w_1)w_1 + (v \cdot w_2)w_2 + \cdots + (v \cdot w_p)w_p$$

Prove that T is a linear transformation.

(2) Let A and B be similar matrices. Show that if A satisfies the equation $A^3 - 3A + I = 0$, then B also satisfies a similar equation $B^3 - 3B + I = 0$.

5. Synthesis (8 points)

Let x be a vector in R^n with $x^T x = 1$, Show that if $A = I - xx^T$, then $rank(A) < n$.

《SE-103+线性代数》期末试题答案(B)

1. Fill in the blank（5×4=20 Pts）

(1) $(2a_0 + a_1) + (3a_1 - a_2)t + (a_0 - 2a_1 + 4a_2)t^2$

(2) **3, Yes**

(3) $A = PDP^{-1} = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$

(4) $a \neq 0$ **and** $a \neq 3$

(5) $[x_1 \ x_2 \ x_3] \begin{bmatrix} 3 & -1/2 & 0 \\ -1/2 & 5 & 4 \\ 0 & 4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

2. Make each statement True or False, and descript your reasons.（5×4=20 Pts）

(1) F **(2) T** **(3) T** **(4) F** **(5) F**

3. Calculation (5×8=40 Pts）

(1) Solution **Since** $B = [b_1 + b_2 + b_3 \quad b_1 + 2b_2 + 4b_3 \quad b_1 + 3b_2 + 9b_3]$

$$= [b_1 \ \ b_2 \ \ b_3] \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix},$$

Then $\det B = \det([b_1 \ \ b_2 \ \ b_3] \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix})$

By the multiplicative property of determinants, $\det B = \det A \cdot \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{vmatrix} = 1 \times 2 = 2$

(2) Solution $\det (A - \lambda I) = (\lambda - 2)(\lambda - 1)$ **. The eigenvalues are 2 and 1, and the corresponding eigenvectors are** $v_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ **and** $v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ **. Next, form** $P = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$ **,**

$D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ **,and** $P^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$

Since $A = PDP^{-1}$**,** $A^6 = PD^6P^{-1} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2^6 & 0 \\ 0 & 1^6 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 190 & -189 \\ 126 & -125 \end{bmatrix}$

(3) Solution A typical element of H **can be written as**

$$\begin{bmatrix} 3a + 7b - c \\ -5b + 8c - 2d \\ 3d - 4e \\ 5b - 8c - d + 4e \end{bmatrix} = a \begin{bmatrix} 3 \\ 0 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 7 \\ -5 \\ 0 \\ 5 \end{bmatrix} + c \begin{bmatrix} -1 \\ 8 \\ 0 \\ -8 \end{bmatrix} + d \begin{bmatrix} 0 \\ -2 \\ 3 \\ -1 \end{bmatrix} + e \begin{bmatrix} 0 \\ 0 \\ -4 \\ 4 \end{bmatrix}$$

$\mathbf{u_1}$ $\mathbf{u_2}$ $\mathbf{u_3}$ $\mathbf{u_4}$ $\mathbf{u_5}$

a. H **is a vector space because it is the set of all linear combinations of a set of vectors.**

b. Row reduce:

$$\begin{bmatrix} 3 & 7 & -1 & 0 & 0 \\ 0 & -5 & 8 & -2 & 0 \\ 0 & 0 & 0 & 3 & -4 \\ 0 & 5 & -8 & -1 & 4 \end{bmatrix} \sim \begin{bmatrix} 3 & 7 & -1 & 0 & 0 \\ 0 & -5 & 8 & -2 & 0 \\ 0 & 0 & 0 & 3 & -4 \\ 0 & 0 & 0 & -3 & 4 \end{bmatrix} \sim \begin{bmatrix} 3 & 7 & -1 & 0 & 0 \\ 0 & -5 & 8 & -2 & 0 \\ 0 & 0 & 0 & 3 & -4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The pivot columns are 1,2, and 4. So { $\mathbf{u_1}$ **,** $\mathbf{u_2}$ **,** $\mathbf{u_4}$**} is a basis for** H **.**

(4) Solution (a) Because the columns a_1 **and** a_2 **of** A **are orthogonal, the orthogonal projection of** b **onto** $\text{Col } A$ **is given by**

$$\hat{b} = \frac{b \cdot a_1}{a_1 \cdot a_1} a_1 + \frac{b \cdot a_2}{a_2 \cdot a_2} a_2 = \frac{2}{7} a_1 + \frac{1}{7} a_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

(b) $\hat{x} = \begin{bmatrix} 2/7 \\ 1/7 \end{bmatrix}$ **, since we already know what weights to place on the columns of** A **to produce** $\hat{b}$**.**

(c) $\| b - \hat{b} \| = \sqrt{(4-1)^2 + (-2-1)^2 + (-3)^2} = 3\sqrt{3}$

(5) Solution **Let** $v_1 = x_1$ **and** $v_2 = x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} v_1 = x_2 - \frac{1}{2} v_1 = \begin{bmatrix} 3 \\ 3/2 \\ 3/2 \end{bmatrix}$**. Thus** $\{v_1, v_2\}$ **is an orthogonal basis for** W**.**

Since $\| v_1 \| = \sqrt{30}$ **and** $\| v_2 \| = \sqrt{27/2} = 3\sqrt{6}/2$**, an orthonormal basis for** W **is**

$$\left\{ \frac{v_1}{\| v_1 \|}, \frac{v_2}{\| v_2 \|} \right\} = \left\{ \begin{bmatrix} 2/\sqrt{30} \\ -5/\sqrt{30} \\ 1/\sqrt{30} \end{bmatrix}, \begin{bmatrix} 2/\sqrt{6} \\ 1/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix} \right\}.$$

4. Prove issues（2×6=12 Pts）

(1) Solution **If** u **and** v **in** R^n**, then**

$$T(u+v) = [(u+v)^T w_1]w_1 + [(u+v)^T w_2]w_2 + \cdots + [(u+v)^T w_p]w_p$$

$$= [(u^T + v^T)w_1]w_1 + [(u^T + v^T)w_2]w_2 + \cdots + [(u^T + v^T)w_p]w_p$$

$$= (u^T w_1)w_1 + (u^T w_2)w_2 + \cdots + (u^T w_p)w_p$$

$$+ (v^T w_1)w_1 + (v^T w_2)w_2 + \cdots + (v^T w_p)w_p$$

$$= T(u) + T(v)$$

It can be shown similarly that $T(cu) = cT(u)$ **for each scalar** c**, so** T **is a linear transformation.**

(2) Solution **Since** A **is similar to** B**, there is an invertible matrix** P **such that** $A = PBP^{-1}$**, then** $A^3 - 3A + I = (PBP^{-1})^3 - 3(PBP^{-1}) + I$

$$= PB^3P^{-1} - 3(PBP^{-1}) + I$$

$$= P(B^3 - 3B + I)P^{-1}$$

$$= 0$$

Thus $B^3 - 3B + I = 0$ **because** P **is an invertible matrix.**

5. Synthesis (8 points)

Solution **Since** $x^T x = 1$**, thus** $Ax = (I - xx^T)x = x - xx^T x = x - 1x = 0$ **.**

Next, we know $x \neq 0$ **because** $x^T x = 1$**. Hence the homogeneous equation** $Ax = 0$ **has a nontrivial solution. This shows that the equation has at least one free variable, and** $rank(A) < n$

中山大学软件学院 2013 级软件工程专业(2013学年秋季学期)

《SE-103 线性代数》期末试题(A卷)

(考试形式：开/闭 卷　考试时间：2 小时)

警示

《中山大学授予学士学位工作细则》第六条

考试作弊不授予学士学位

方向：_____　姓名：_____　学号：_____

1. Fill in the blanks（6×4=24 Pts）

(1) Let $A=\begin{bmatrix}1&0&1\\0&2&0\\-2&0&1\end{bmatrix}$ and $A^TB-A-B=I$, where A and B are 3×3 matrices, then $|B|$=_____.

(2) Given a subspace $H=\left\{\begin{bmatrix}a-3b+6c\\5a\\b-2c-d\\0\end{bmatrix}: a,b,c,d \text{ in } R\right\}$, a basis is _____, and the dimension of H is _____.

(3) Let $A=\begin{bmatrix}0&1\\-8&4\end{bmatrix}$ act on C^2. Then an eigenvalues of A is λ=_____. And a basis for the eigenspace corresponding to λ is _____.

(4) Let $y=\begin{bmatrix}5\\-9\\5\end{bmatrix}$ and $W=\text{span}\{u_1,u_2\}$, where $u_1=\begin{bmatrix}-3\\-5\\1\end{bmatrix}$ and $u_2=\begin{bmatrix}-3\\2\\1\end{bmatrix}$, then

$\text{proj}_W y$ = _____, and the distance from y to W is _____.

(5) Let $A=\begin{bmatrix}1 & 5\\ 3 & 1\\ -2 & 4\end{bmatrix}$ and $b=\begin{bmatrix}4\\ -2\\ -3\end{bmatrix}$, then a least-squares solution of $Ax=b$ is _____, and the associated least-squares error is _____.

(6) Let A be the matrix of the quadratic form $(x_1+x_2)^2+(x_2-x_3)^2+(x_3+x_1)^2$, then A = _____, and rank A is _____.

2. Make each statement True or False, and descript your reasons. (6 ×3 = 18 Pts)

(1) If A is a positive definite symmetric $n\times n$ matrix, then A^{-1} is also positive definite.

(2) Suppose a 3×5 matrix A has dim Row A =3. Then the equation $Ax=b$ always has a unique solution.

(3) If V is a vector space having dimension n, and if S is a subset of V with n vectors, then S is linearly independent if and only if S spans V.

(4) Two distinct eigenvectors corresponding to the same eigenvalue are always linearly dependent.

(5) If A is produced by multiplying row 2 of B by 3, then det A =3det B.

(6) If a matrix U has orthonormal columns, then $UU^T=I$, where I is the $n\times n$ identity matrix.

3. Calculation (5 ×8 = 40 Pts)

(1) Let $A=\begin{bmatrix}1 & 0\\ -1 & 2\\ 0 & 1\end{bmatrix}$, $B=A^TA$

a. Find the eigenvalues and the corresponding eigenvectors of B.

b. Computer B^k, where k represents an arbitrary positive integer.

(2) Find a QR factorization of $A=\begin{bmatrix}5 & 9\\ 1 & 7\\ -3 & -5\\ 1 & 5\end{bmatrix}$.

(3) If $B=\begin{bmatrix}2 & 3 & 0\\ 3 & 1 & 1\\ 0 & 1 & 0\end{bmatrix}$, and if A and B are similar.

a. Find $|A-2I|$.

b. Let $\vec{x}=\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$. Computer x^TBx for the matrix B.

(4) Suppose $A=\begin{bmatrix}1 & 0 & -2 & 1\\ 0 & 1 & a & 0\\ 2 & 1 & 0 & a\\ 0 & 0 & 0 & a-1\end{bmatrix}$. If the columns of the matrix A are linearly dependent and $a\neq 1$.

a. Find a.

b. Find bases for the null space , the column space, and the row space of the matrix A.

(5) If T is the linear transformation from P_2 to P_2.

a. Suppose the set $B=\{1+t,1+t^2,t+t^2\}$ is a basis for P_2. Find the coordinate vector of $p(t)=1+3t-2t^2$ relative to B.

b. IF $T(a_0+a_1t+a_2t^2)=2a_1+4a_2t^2$, find the C-matrix for T, when C is the basis $\{1,\ t\ ,t^2\}$.

4. Prove issue（2 ×9 = 18 Pts）

(1) Suppose $A=I-3uu^T$, where u is a unit vector in R^n and I is the $n\times n$ identity matrix.

a. Let u be an eigenvector of A, find the corresponding eigenvalue.

b. If v is any vector orthogonal to u, show that v is an eigenvector of A and find the eigenvalue.

(2) Suppose A and B are both $n\times n$ symmetric matrices. Show that AB is an symmetric matrix if and only if $AB=BA$.